Legibility Is Not Deployability: Four Gates for LLM-Assisted Prediction-Market Trading

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May 2026

Abstract

Prediction markets invite a seductive story about LLM agents. Contract rules are written in natural language, public prices provide immediate targets, order books are inspectable, and news can be summarized inside a context window. A model can read a market, generate a probability, and appear one comparison away from a trade. This paper studies where that story breaks. We analyze a failure-preserving Abel AI Lab corpus covering 13 active campaigns and 156 instrumented adversarial-review rounds from 2026-Q2, preserving 225 recovered route decisions, 52 strict attack rows, and 40 quote-bank rows rather than only successful trajectories. Across the corpus, apparent LLM-accessible edges repeatedly failed at four operational gates: **sourceability**, **fillability**, **capacity**, and **anchor stability**. Mechanical price relationships were often real historically but closed or too small in current books; LLM extraction systems produced high-confidence fields without source-grounded discriminators; visible spreads collapsed under executable depth, fees, settlement, and bankroll utilization; and LLM priors near market consensus required anchor controls before being treated as independent signals. We formalize **ceiling-pin layer migration**: tightening a rubric can move an LLM-extraction defect from the scoring layer to the prompt/output layer unless verification leaves the model loop and checks the external object. Across 11 trading-focused campaigns, **zero routes** cleared all four gates strongly enough to support a durable greater-than-$1k/month retail deployment claim. The contribution is neither a trading strategy nor a market-efficiency theorem; it is a boundary map of where fluent market reasoning loses the right to be called a trade, and a failure-preserving corpus that lets others test their own boundaries.

1 Introduction

Prediction markets [Wolfers and Zitzewitz, 2004] look almost tailor-made for language-model agents. A market title names a future event. A rule page describes settlement. Public prices expose the crowd's current view. News and filings supply context. From this interface, the agentic workflow writes itself: read the market, infer a probability, compare that probability with price, and trade the gap.

That workflow fails at the word *trade*. Fluent market descriptions are cheap; deployable routes are not. A route can be readable without being sourceable, visible without being fillable, profitable in percentage terms without carrying capacity, and numerically precise while still inheriting a prompt-visible consensus anchor. Legibility is the model-facing layer. Deployability begins only when external sources, executable books, capacity arithmetic, and prior-independence checks all agree.

This paper turns that gap into an empirical object. Rather than presenting a winning strategy, we analyze a failure-preserving archive from Abel AI Lab in which LLM-assisted research loops repeatedly tried to convert prediction-market observations into retail-scale trading routes. The campaigns tested threshold ladders, duplicated-market relationships, date-ladder monotonicity, cross-venue spreads, market-internal residuals, LLM narrative priors, event-contract rule extraction, bankruptcy-claim sourcing, and SEC special-situation extraction. Crucially, the archive keeps the discarded paths: proposed, selected, rejected, deferred, promoted, and terminated routes remain in the denominator.

Preserving that denominator matters because both trading research and agent research are structurally prone to survivor stories. Backtests that once worked remain memorable long after live markets adapt [Brown et al., 1992, Bailey and Lopez de Prado, 2014]. Agents that reach a benchmark score leave cleaner artifacts than routes killed by peer attack before implementation. In the corpus studied here, those killed routes are the signal. They reveal where alpha-shaped objects lost the right to be called trades.

The public artifact contains 17 raw campaign directories and 164 raw round directories. Four sparse legacy directories are retained for inventory honesty but excluded from mechanism claims. The active corpus contains 13 campaigns and 156 instrumented rounds. Of those rounds, 97 are core infrastructure hardening from live-readiness and websocket/cron-alignment campaigns. The substantive trading subset contains 59 trading-focused rounds across 11 active trading campaigns. This partition prevents safety and measurement work from being mislabeled as trading edge.

Across that record, four gates repeatedly separate model-readable routes from deployable routes:

1. **Sourceability.** The trade-weighted claim must be verified by an external object: a rule clause, filing span, book snapshot, timestamped row, or realized ledger entry.
2. **Fillability.** A quoted price must be executable at the intended size and time, not merely visible in a UI or historical table.
3. **Capacity.** The route must survive fees, slippage, settlement, holding period, opportunity frequency, and bankroll utilization.
4. **Anchor stability.** An LLM prior must remain materially stable when market consensus is hidden, revealed, or perturbed before it can be treated as independent signal.

Our claim is intentionally bounded. In this single-lab, single-period corpus, no analyzed LLM-assisted prediction-market route cleared all four gates strongly enough to support a durable greater-than-$1k/month retail deployment claim. That does not imply prediction-market efficiency, LLM trading impossibility, or a universal market boundary. It shows a repeatable operational pattern: the hard failures occurred after language made the route seem available.

1.1 Contributions

First, we release **a failure-preserving empirical corpus**—to our knowledge the first publicly released artifact of this kind for LLM-assisted retail prediction-market research—with rejected, deferred, and promoted routes preserved alongside the surviving ones, so that external reviewers can audit failure modes rather than only successful trajectories. Second, we distill **four operational gates**—sourceability, fillability, capacity, and anchor stability—that repeatedly erased apparent LLM-accessible edges, and provide a gate codebook (Table 2) that names each gate's passing condition and its typical false-green failure. Third, we formalize **ceiling-pin layer migration**, a cross-mechanism failure mode in which extraction confidence survives a rubric repair unless the checker leaves the model output and verifies the source object itself; this is the paper's central novel apparatus and the conceptual export beyond trading. Fourth, we document an adversarial research protocol, the **R0 program gate**, that made launch ambiguity countable before implementation burn—an LLM-trading-research analogue of preregistration [Nosek et al., 2018] adapted to the speed of agent loops, where a single afternoon can otherwise burn ten launch attempts before the first ambiguous claim surfaces.

Table 1: Failure-preserving corpus card. The denominator is retained before claims narrow. The bottom row is the headline: across 11 trading-focused campaigns and 59 trading rounds, zero routes cleared all four gates strongly enough to support a durable greater-than-$1k/month retail deployment claim.

Artifact or partition	Count
Raw campaign directories	17
Active campaigns	13
Trading-focused campaigns	11
Instrumented rounds	156
Recovered route decisions	225
Strict attack rows	52
Quote-bank rows	40
Durable deployment routes	**0**

Figure 1: Four operational gates. A model-readable route earns trading language only after source verification, executable depth, capacity-weighted economics, and anchor-stable prior elicitation.

2 Positioning

Forecasting benchmarks ask whether models can estimate unresolved events [Karger et al., 2024, Schoenegger et al., 2024]. Live-market benchmarks ask whether agents can act in dynamic financial environments [Cheng et al., 2026]. Prediction-market microstructure studies ask how prices, depth, latency, and information interact [Wolfers and Zitzewitz, 2004, Manski, 2006]. Our question sits between those strands: what breaks when an LLM-generated forecast is promoted into a retail trading route?

Answering that question requires keeping failures that standard scoreboards often discard. A leaderboard can report final performance while hiding route attrition. A strategy note can celebrate a surviving mechanism while omitting the variants that died at source, fill, or capacity gates. A simulation can finish with a model comparison table even if the most important live-market defect would have appeared before execution. We instead treat the route ledger as the unit of evidence.

Anchoring and sycophancy research supplies one part of the failure map [Lou and Sun, 2024]. If an LLM sees market consensus before generating a prior, the output can blend independent evidence with prompt-visible consensus. The sister MarketAnchor artifact measured this problem in a stored prediction-market forecasting run: mean net signed prior drift was 256.73 probability basis points over 75 matched triplets. We use that result only as a hygiene warning. Prior drift is not expected return, not calibration error, and not realized profit. It becomes operationally relevant because a trading system may accidentally treat a quote-conditioned output as an independent signal.

Financial-LLM work supplies the positive backdrop. Models can summarize filings, parse market text, search news, generate hypotheses, and accelerate analyst workflows [Schoenegger et al., 2024]. Nothing in this paper denies that utility. Our negative result is narrower and more useful: a trade inherits an LLM output

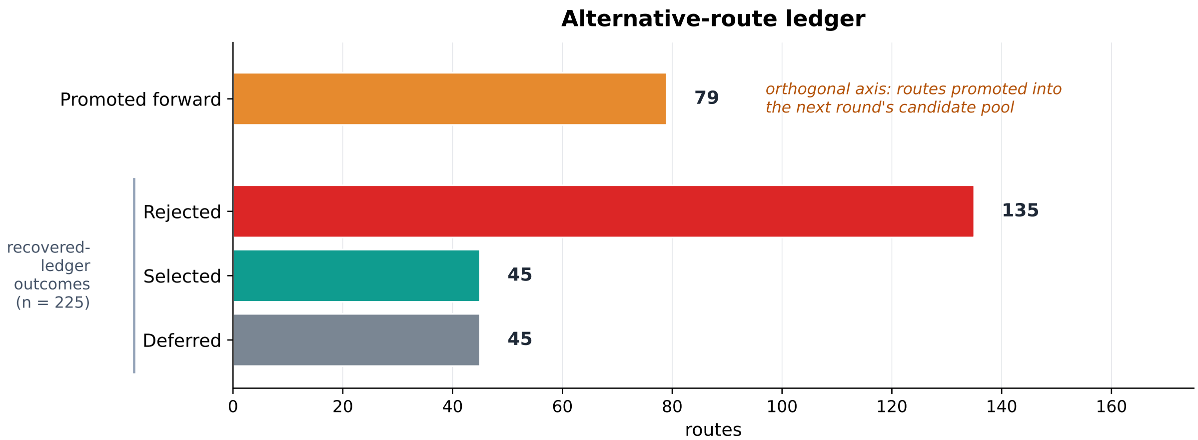

Figure 2: Route attrition ledger. The lower three bars are terminal outcomes for the 225 recovered route decisions (135 rejected, 45 selected, 45 deferred). The top bar tracks an orthogonal axis: 79 routes whose `promoted_to_next_round` flag is true, i.e. carried forward into the next round's candidate pool. Promotion and selection are tracked separately because a route can be advanced for further attack without becoming the chosen implementation path. The missing denominator is the route that looked plausible, was attacked, and failed before capital could scale.

only after non-LLM gates verify the object that carries risk.

3 Corpus and Audit Method

Our evidence base is small by lab count and dense by audit trail. Each active campaign records program state, route alternatives, selected routes, metric movement, peer attacks, blocker status, and round-level artifacts. The archive is not a random sample of all possible strategies. It is a preserved trace of one lab's repeated attempts to turn LLM-accessible market information into retail deployment candidates.

Raw inventory comes first. Seventeen campaign directories and 164 raw round directories remain in the repository. Four legacy directories contain sparse or empty artifacts; they remain in inventory counts but do not support mechanism claims. Thirteen campaigns form the active corpus. Twelve have full state JSON plus peer artifacts, and one is peer-files-only. This boundary prevents thin directories from masquerading as instrumented evidence.

Round counts are partitioned before analysis. Infrastructure campaigns contributed 97 rounds, dominated by live-readiness and websocket/cron-alignment hardening. Trading-focused claims use 59 rounds across 11 active trading campaigns. This separation is not cosmetic. A system can make enormous engineering progress while producing no deployable trading evidence; conflating those categories is one route to false confidence.

Route decisions provide the failure denominator. The recovered registry contains 225 route rows: 45 selected, 135 rejected, and 45 deferred. Promotion is tracked separately because a route can be carried forward without becoming the chosen implementation path; the per-row `promoted_to_next_round` flag is true on 79 routes. In other words, the archive preserves search discipline, not just search success.

Attack rows make objections auditable. The strict attack registry contains 52 peer attacks after a noisy earlier parse was quarantined. A separate methodology-self-attack file keeps manuscript-construction objections out of cross-campaign trading recurrence counts. The quote bank contains 40 verbatim evidence rows with per-campaign and peer-role caps, so one dense infrastructure campaign cannot dominate the evidence base.

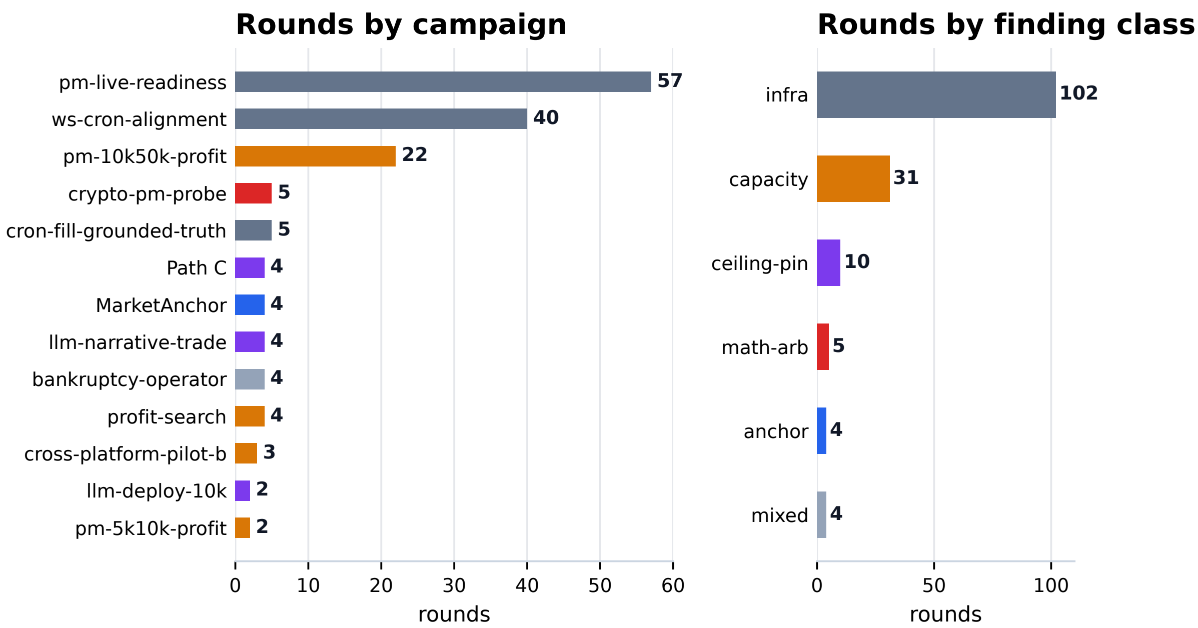

Figure 3: Campaign-level audit trail. Rounds are separated by campaign and by dominant finding class so measurement hardening is not misreported as trading edge.

4 Gates I and II: Sourceability and Fillability

Public, mechanical relationships were the first routes to face the source and fill gates. Several mechanisms had real historical appeal. A threshold-ladder arsenal recorded 327 walk-forward trades over 14 months at positive per-leg returns. Yet a current-market query on 2026-05-03 found zero violations across 4,365 live markets in the relevant ladder families. Historical validity did not create current sourceability.

Market-making failed through a different channel. The websocket/cron-alignment campaign tested spread-capture settings against live fill and markout. At 100 bps, 5 of 6 fills lost money; at 200 bps, 0 of 3 were winning; at 300 bps, the sample was too small to rescue the route. Visible spread capture was not enough because adverse selection and markout dominated the apparent maker edge—the canonical microstructure result that informed inventory pricing pre-LLM [Glosten and Milgrom, 1985] reappears whenever an agentic maker quotes into a flow it does not characterize.

Cross-venue ideas exposed semantic fillability. A spread across platforms is not one trade unless the event definitions match, both venues are accessible, fees and settlement timing are known, and both books carry executable depth. The appeal of cross-venue arbitrage came from legibility: two prices could be read and compared. The blocker came from deployability: shared semantics and synchronized execution remained unproven.

Internal-feature routes reached a parallel boundary. On liquid Polymarket markets, models using only market-internal price, volume, momentum, and category features underperformed the market snapshot probability. Public features visible to other participants did not become private signal by passing through a model.

Sourceability and fillability therefore come before strategy language. A candidate route begins with live violation counts, timestamp alignment, settlement-rule grounding, and order-book depth at the decision time. Backtest memory can motivate a probe; it cannot substitute for current executable evidence.

Table 2: Gate codebook. Each gate names a passing condition and a common false-green failure.

Gate	Passing condition	Typical false green
Sourceability	A trade-weighted claim is tied to an external object: filing span, rule clause, book snapshot, timestamped row, or ledger entry.	A model output looks grounded but the source span is absent, ambiguous, or mismatched.
Fillability	The quoted edge is executable at the required size and time.	A spread is visible, but depth, timing, venue access, or settlement mechanics make it non-executable.
Capacity	Net economics survive fees, slippage, hold time, opportunity frequency, and bankroll utilization.	A large percentage edge exists only on trivial dollars or on shadow capacity that never reaches a live ledger.
Anchor stability	The prior remains stable when consensus is hidden, revealed, or perturbed.	A quote-visible prior is treated as independent despite prompt contamination or ensemble instability.

Panel	n	win	cum_pnl	sharpe*	gate-failure attribution
pmDup	99	100%	+$29.0	17.43	G2 – 100% win is paper-math idealization
pmSpread	131	87%	+$41.7	9.27	G2 – partial-fill / settlement uncosted
pmMlSpread	78	100%	+$19.1	7.08	G2 – same idealization as pmDup
pmLadder-postfix	41	—	+$14.1	11.84	G1 – retroactive fix on a closed alpha class
pmLadder original	50	—	+$1.6	0.28	G1 – 0/4365 violations 2026-05-03 (§4)
pmWdw	131	16.8%	−**$0.41**	−**0.68**	mechanism *fails* under realistic execution
pmByDate	5	100%	+$0.9	1.16	G3 – n=5; “live alpha” claim has 5 obs
pmOuPlayer	0	—	0	—	panel unstarted on snapshot date

Table 3: Eight paper-trade panel snapshot, 2026-05-07. Source: `data/anc/dashboard-snapshot-2026-05-07.json`. * Per-trade Sharpe $\text{mean(pnl)}/\text{std(pnl)} \times \sqrt{n}$. Not annualized; monotone in n for any positive-EV mechanism by construction. The dashboard tooltip flags this directly: a rising Sharpe is the formula, not the alpha. Mechanism descriptions are deferred to Appendix B.

4.1 Worked examples: eight paper-trade panels

The trading-internal dashboard runs eight paper-trade panels, each instantiating a candidate math-arb mechanism against the live Polymarket universe. The panels share an idealized executor model—best-of-book quote at detection time, no slippage, no partial fills, no settlement disputes—and write resolved trades to a shared state file every five minutes. Table 3 reports the snapshot of 2026-05-07. Paper PnL is not fabricated: it is the executor model applied to actually resolved Polymarket events. But `live_mode_active = False`, `broker_mode = DRYRUN`, and `ready_to_flip = False` across the full operator stack. Zero trades have been executed against a real broker on any of the eight panels.

Seven of the eight panels report monotonically rising Sharpes. Only one—pmWdw, the 3-way W/D/W MECE basket—reports realistic execution honestly: a 16.8% win rate, −$0.41 cumulative paper PnL, and a Sharpe of −0.68. The contrast is the four-gate map in miniature. Idealized executor models can manufacture arbitrarily attractive Sharpes for any positive-EV math-arb mechanism, but the moment slippage, partial fills, and settlement disputes are priced in (pmWdw) the mechanism either fails outright or stalls at execution depth (pmDup, pmSpread, pmMlSpread) before live capital can be committed. Two of the eight panels (pmLadder, pmLadder-postfix) operate inside an alpha class the project itself has classed as closed. One (pmByDate) reports a paper record of 5 trades against an EV-target label of $55k. None of the eight has cleared the four operational gates strongly enough to support a live-deployment claim, and the operator-readiness panel does not assert that any of them should.

The exhibit is the paper's title in numbers. The seven idealized panels are *legible*—well-defined arb formulations, clean per-trade Sharpe math, automated paper executor—and *not deployable*. The one panel whose executor models adverse selection at all (pmWdw) reports the mechanism failing. The asymmetry—large hero numbers above the fold, formula caveat under hover, zero live fills under any panel—is itself an instance of the legibility-vs-deployability gap this paper documents.

5 Ceiling-Pin Layer Migration

False high confidence was more dangerous than low confidence. We call the recurring pattern *ceiling-pin layer migration*.

Definition 1: Ceiling-pin. A scoring or gating variable is ceiling-pinned when it reaches its maximum or near-maximum value across many candidate rows while failing to discriminate trade-relevant correctness.

Definition 2: Layer migration. A ceiling-pin defect migrates when a repair at one layer, such as a rubric or parser constraint, removes the original failure surface but causes an analogous failure to reappear at another layer.

Definition 3: Ceiling-pin layer migration. A tightened LLM-extraction gate exhibits ceiling-pin layer migration when it no longer accepts empty or generic fields, yet still accepts ungrounded answer-shaped outputs that satisfy the new checker without being verified against the external source object.

The cleanest rubric-side case is `llm-deploy-10k`. A rule-reader confidence score awarded 25 points each for resolution source, deadline, fallback clauses, and expected payoff table. Polymarket Gamma metadata made those fields easy to populate, so every sampled event cleared the rule-reader threshold. The score looked strict and became non-discriminative. A high confidence number then coexisted with a negative paper-trade result.

Path C, the SEC special-situations probe, reproduced the defect in a new domain. R1 accepted truthy strings for regulatory and consideration axes. R2 tightened the rubric, but the failure moved: the model generated plausible allowlisted phrases such as "none required" even when the cited filing chunks did not contain them. The checker still evaluated the model's output rather than the filing.

Narrative-prior trading supplies the adjacent prompt-gate case. A prior route was supposed to read sources first, emit an independent probability, and only then compare with market price. One branch exposed the market quote before prior generation. Later repairs masked some anchor exposure, but source grounding, semantic agreement, ensemble variance, realized PnL, and capacity evidence remained unresolved. The route was not a completed profit result; it was a warning that prompt order can masquerade as timestamp order.

What links the three cases is not a single bug but a shared assumption: that a check on the model's output is the same as a check on the source object. It is not. The rubric layer can be tightened, the prompt-gate order can be repaired, and the confidence number can climb, all while the underlying ground truth is never re-examined. Ceiling-pin layer migration is the structural reason a stricter checker can become a weaker one, and the only stable repair is to verify outside the model loop.

6 Gate III: Capacity

A 23.6% spread on $5.34 is not a retail trading strategy. Capacity is the point where percentage edge becomes dollars, utilization, and risk.

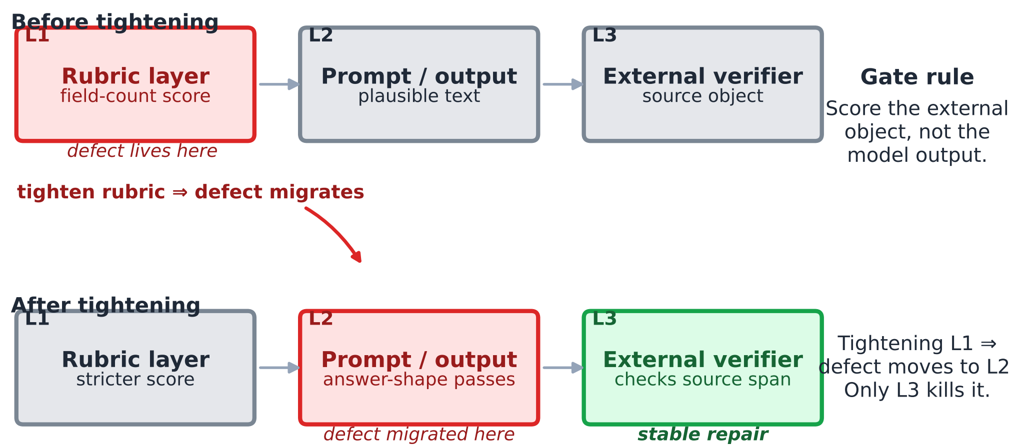

Figure 4: Ceiling-pin layer migration. Tightening a rubric can move the defect from field-count scoring to prompt-side plausible text unless verification leaves the model and checks the source object.

We use \$1k/month as an operational retail-viability threshold, not a formal statistical definition of profitability. Routes below that floor can still be scientifically interesting, but they do not justify deployment language unless opportunity frequency, holding period, variance, and monitoring overhead all work at the intended bankroll.

A candidate row can be written schematically as

$$E_j = \min(C_j, B_j)\big(g_j - f_j - s_j - \varepsilon_j\big),$$

where C_j is executable capacity, B_j is budget, g_j is gross edge, f_j is fees, s_j is slippage, and ε_j is an error buffer. The paper does not estimate E_j for every row. Instead, it treats the inputs as gates: missing capacity, missing fee information, or missing execution timing is enough to block a deployment claim. The three examples below illustrate three different missing-input failures.

Crypto-prediction-market probes show the tiny-cycle failure: C_j collapses. A gross spread looked attractive, but the bottleneck side of the book carried only about \$5.34 of fillable NO depth. LLM narrative priors produced a larger but still constrained example: one sharp row had roughly \$200-\$500 immediately deployable before slippage, and daily deployability required many independent rows—small C_j multiplied by uncertain row frequency. Bankruptcy-claim sourcing failed through a different channel: C_j as visible book depth was not the binding constraint; access depth was. Public scrape evidence did not reveal enough class-specific executable inventory.

The most constructive counterexample is the pm-10k50k-profit campaign. It verified \$85,446.78 of cumulative-to-50bps candidate capacity across four accepted candidates, but kept that frontier shadow-only because current evidence, priors, and paper execution had not yet been attached. That discipline is the point. Capacity measured without evidence is not PnL; capacity omitted from evaluation is phantom scale.

7 Gate IV: Anchor Stability

An LLM prior near a market price is not independent simply because the row calls it a prior. Anchor stability is therefore a trading gate, not just a benchmark curiosity. The phenomenon has experimental backing in the broader anchoring-bias literature on language models [Lou and Sun, 2024], where models adjusted estimates

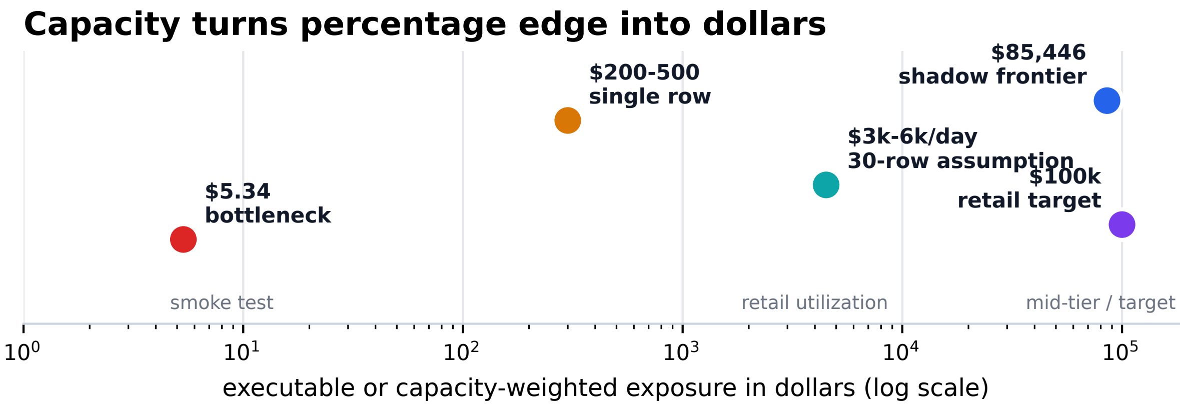

Figure 5: Retail capacity ladder. Visible spread is not deployable capacity: a route must survive fees, depth, holding time, and bankroll utilization.

toward arbitrary numerical anchors even when the anchor was unrelated to the target quantity; for trading, the question is whether the same effect is large enough at retail-edge magnitudes to invalidate a flagged route.

MarketAnchor v0.1.2 measured consensus-anchor sensitivity in one stored prediction-market forecasting run. It reported mean net signed drift of 256.73 probability basis points across 75 complete triplets over eight contracts. That number is a probability-prior movement, not expected return, not realized profit, and not forecast accuracy. We use it only for a narrow operational rule: if a prompt exposes consensus before eliciting the prior, the output may be contaminated by the comparison signal.

For trading systems, the buffer should be conservative. When an apparent LLM-prior edge is below roughly 500 bps and the prior was elicited near market price or consensus context, the edge is suspect unless anchor controls are applied. The 500 bps buffer is not a universal profit cutoff; it is a hygiene threshold above observed anchor drift and campaign-specific ablation movements.

Narrative-prior campaigns show why semantic order matters. A row schema can label the prior timestamp as earlier than the price comparison, while the model has already seen the market quote inside the prompt. Robust LLM-prior routes should make the no-quote prior canonical, hide consensus until after prior generation, log source publication dates, measure ensemble variance, and attach realized exit marks before crediting a flagged row.

8 Abelian Adversarial Methodology

Method is part of the result because route failure had to be made visible. The Abelian loop records program contracts, route alternatives, selected routes, rejected routes, metric deltas, peer attacks, blocker status, frame breaks, and commit-gated progress. That state machine turns a failed route from anecdote into evidence, and adapts the preregistration discipline that has matured in the empirical sciences [Nosek et al., 2018, Munafò et al., 2017] to the speed of agent-driven research. Classical preregistration is measured in weeks; an R0 program gate runs in minutes, because the cost it controls is no longer human study time but agent-loop launch burn. The transferable insight is that preregistration is fundamentally about contracting on what would count as evidence before the search begins, and that contract is even more valuable when the search itself is cheap.

The most important step is the pre-launch R0 program gate. Before implementation begins, a program specifies target, eval, metric, budget, file scope, data gates, and stop conditions. A baseline eval runs before mutation. A program-peer challenge attacks the contract before R1 starts. The contract hash and human

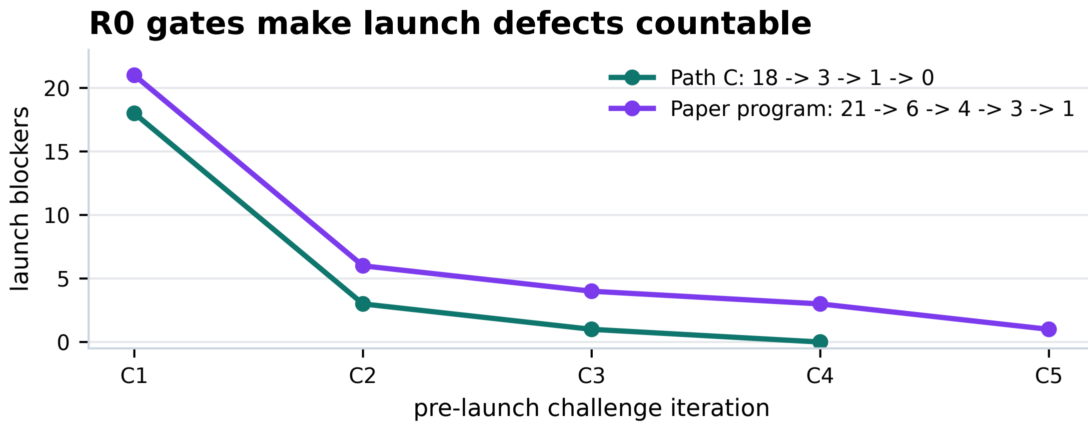

Figure 6: R0 program-gate convergence. Pre-launch adversarial review makes launch ambiguity countable before implementation burn.

confirmation define the launch boundary. The R0 gate is the cheap point at which a fuzzy goal can be sharpened into a measurable claim, the analogue of a pre-analysis plan for a trading-research loop that may otherwise produce many uncomparable mutations.

R0 does not guarantee success. Its value is cheaper failure. In Path C, the R0 sequence narrowed 18 launch attacks to 3, then 1, then 0 before implementation. In the manuscript-building program, the sequence narrowed 21 attacks to 6, then 4, then 3, then 1 before extraction. Those contractions do not prove correctness. They show that ambiguity, borrowed authority, scope drift, and evidence-definition defects were counted while they were still cheap to fix.

Asymmetric peer roles expose different surfaces. One peer attacks framing, scope, reviewer anticipation, and strategic plausibility. Another attacks extraction, transcript mining, recomputation, schema validity, and exact evidence behavior. The loop advances only when both kinds of objections have been addressed or explicitly deferred. This asymmetric red-teaming is closer in spirit to model-written-evaluations work [Perez et al., 2022] than to consensus review: convergence is not the goal, attack-exhaustion is.

9 Integrated Findings

A single chain connects the four gates. A route may have historical evidence but no current violation. It may have a visible spread but no executable depth. It may have depth but too little capacity. It may have a model prior but no anchor independence. It may have a rule-reader confidence score but no source-grounded discriminator. Failure rarely arrived as one dramatic crash; it appeared as missing evidence at the boundary between language and execution.

For builders, the ordering is practical:

1. Start with the external source object, not the model output.
2. Verify executable fillability at the intended size and timestamp.
3. Compute capacity before calling an edge deployable.
4. Elicit LLM priors under anchor-masked conditions before comparing to market price.
5. Preserve rejected and deferred routes so search does not become a winner-only narrative.

LLMs still have a large constructive role. They can accelerate source retrieval, rule summarization, anomaly triage, hypothesis generation, code inspection, and adversarial review. The negative result is not that

LLMs have no market value. It is that the trade inherits the LLM output only after non-LLM gates verify the source and execution facts.

The actionable framing for the LLM-trading research community is inversion: instead of asking "where does our model produce alpha?", ask "where do our four gates fail?", and route engineering effort to the weakest gate first. A team that is honest about which of C_j, f_j, s_j, source span, or anchor exposure is currently missing has a shorter path to a tradable route than a team chasing a higher benchmark score.

10 Limitations

This study is single-lab, single-period, and largely single-operator. Campaigns were not randomly sampled from all possible prediction-market strategies. They reflect Abel AI Lab's research preferences, access constraints, tooling stack, and 2026-Q2 market conditions. The corpus therefore acts as a non-random witness rather than a population estimate; absent a multi-lab replication, the standard caveats about backtest overfitting and selection effects [Bailey and Lopez de Prado, 2014] apply to mechanism interpretation, not just to numerical performance.

Ceiling-pin evidence remains suggestive. The strict extraction/gating count is n=2, and the broader prompt-gate family is n=3. This supports an engineering warning, not a universal law about all LLM systems.

Several campaigns predate the mature R0 gate. Their artifacts still support route-level mechanism claims when evidence is sourceable, but they do not support a claim that the entire corpus was prospectively designed under the final methodology.

Capacity estimates are heterogeneous. Some routes measured book depth, some measured candidate capacity, and others failed because access depth was missing. The paper therefore treats capacity as a gate rather than a fully normalized expected-value model.

External reproducibility depends on the public repository and ancillary files. The key corpus registries are public, but external reviewers may still need raw prompt logs, response logs, market-data snapshots, and line-level source artifacts for deeper replication.

11 Open Problems

Anchor-controlled priors should be tested prospectively. A follow-on study should elicit hidden, visible, counterfactual, and source-grounded priors, then report no-quote priors, with-quote drift, ensemble variance, and realized after-fee PnL separately.

Source-grounded extraction needs model-independent verification. A filing or rule field should be accepted only when the system returns the source span and a verifier can check the span before confidence is scored.

Participant behavior may offer stronger signals than model narration. Insider transactions, whale flows, unusual options volume, borrow-rate moves, or order-flow imbalance differ from LLM priors because the information source is behavior rather than prose. Those routes require timestamped leak checks and capacity-weighted deployment tests from the start.

Multi-model replication remains open. Base models, open-weight systems, domain-tuned finance models, and deterministic API settings may amplify or reduce ceiling-pin and anchoring failures. Moving from a boundary map to a general claim requires that replication.

12 Conclusion

A model-readable route is not a tradable route. Prediction markets make LLM agents feel close to capital allocation because the interface is language. This corpus points in the opposite direction: language is the easy layer. Deployability begins only after the route survives source verification, executable depth, capacity-weighted economics, and anchor-stable prior elicitation.

Failure is therefore not empty. In LLM-assisted trading research, the rejected route is the missing denominator. Preserved failures show where apparent edges lose their source, fill, capacity, or independence. That is the paper's central contribution: not a strategy, but a map of the boundary between fluent market reasoning and deployable market action, and a public corpus other labs can replicate against. The boundary will move; the four gates are the coordinate system for measuring how far.

Appendix B: Mechanism Descriptions

Each panel in § Worked Examples implements a candidate math-arbitrage mechanism on Polymarket. This appendix documents, for each of the eight, (i) the proposition family, (ii) the violation rule that triggers paper entry, (iii) the idealized executor model that produces the dashboard PnL, and (iv) the realistic-execution gap that explains each panel's gate-failure attribution in Table 3.

B.1 pmLadder — Ordered-threshold ladder. Polymarket lists tennis match O/U (e.g., "match games O/U 21.5"), set 1 games O/U, total sets, and crypto-price thresholds (e.g., "BTC reach $150k by Dec 31"). For two adjacent thresholds $X < Y$ in the same contract family, monotonicity requires $P(\geq X) \geq P(\geq Y)$. A violation—hard ask plus easy bid implying the inverse—is the entry signal. The paper executor takes both legs at top-of-book at detection. Realistic execution faces multi-group thousand separators in the question text, LOW/HIGH modifier flips, FDV pre-launch markets with stale pricing, and the `groupItemThreshold` field, which is an MECE-sibling ordinal index rather than a monotone threshold and produces structural false positives if used as one. Alpha class **closed** 2026-05 after a 327-trade arsenal saw 0 violations across 4 365 live markets (§4).

B.2 pmLadder-postfix — Retroactive cross-deadline fix. Earlier ladder runs collapsed cross-deadline duplicates ("hit $150k by June 30" and "hit $150k by Dec 31") into a single ladder, surfacing spurious violations. The postfix variant disambiguates by adding the deadline string to the ladder group key. Sharpe 11.84 on 41 paper trades is the *paper* improvement; the underlying alpha class remains closed, so the panel persists for archive completeness rather than as a deployment candidate.

B.3 pmDup — Cross-event duplicate proposition. The same proposition appears in two distinct events with different prices: e.g., "Will Israel strike Yemen this month" priced in two parallel events. A violation is a tier-threshold price gap at matched semantics. The paper executor assumes both books fill at observed top-of-book; realistic execution faces semantic-equivalence disputes, asynchronous resolution, and partial cancellation when one event resolves before the other. 99 paper trades, 100% win, Sharpe 17.43 is the idealized math; under any non-zero slippage assumption, the realised win rate falls into the same regime as pmWdw.

B.4 pmSpread — Sport line monotonicity. Sport spread markets at adjacent points (e.g., "Lakers -5.5" and "Lakers -6.5") satisfy a monotonicity that fails in roughly 1% of NBA / NFL listings on Polymarket. The arsenal-projected EV target is $160k/yr; the paper executor again models top-of-book fills at observation time on both legs without partial-fill or settlement-cost charges.

B.5 pmMlSpread — Moneyline × spread cross-family. For a given game the moneyline price and the spread price imply a joint constraint via point-spread-to-win-probability translation. A violation is when the implied win probability from one leg contradicts the other beyond a tier threshold. Restricted to NBA + NFL + MLB; other sports lack parallel ML+spread liquidity. 78 paper trades, 100% win, Sharpe 7.08—the same paper-fantasy class as pmDup.

B.6 pmWdw — 3-way W/D/W MECE basket (the honest panel). For a soccer match, Polymarket lists three outcomes—Home, Draw, Away—priced as a MECE basket: $P(H)+P(D)+P(A)$ should sum to 1. A violation has $\sum p \neq 1$ above a tier threshold. The paper executor takes all three legs at top-of-book; the realistic executor models adverse selection by charging the leg most exposed to settlement risk. 131 paper trades, 16.8% win rate, $-\$0.41$ cumulative, Sharpe -0.68. **This is the only panel whose realistic execution model says the mechanism does not work.** Its honesty under adversarial pricing is the negative space against which the seven remaining panels' idealized Sharpes should be read.

B.7 pmByDate — Multi-deadline same-proposition. The same proposition with different end dates ("hit \$150k by June 30" versus "hit \$150k by Dec 31") has an implicit ordering: longer deadlines should price at least as high. A codex-pilot backtest reported 154 pairs with median violation 18.6% and a modeled EV of \$55k. The live panel has resolved **5** paper trades. Internal memory labels multi-deadline LP arb as the project's "only live alpha"; the five-observation paper record does not yet support that claim at any conventional power threshold.

B.8 pmOuPlayer — Per-player NBA O/U monotone. Per-NBA-player O/U props at adjacent thresholds (e.g., "LeBron points O/U 25.5" versus "O/U 26.5") satisfy a monotonicity analogous to pmLadder. Backtest projection is \$20k EV. The panel was **unstarted** on the snapshot date: zero resolved paper trades, scanner not yet live.

The eight panels share two structural properties. *First*, seven of the eight use idealized executors that price in zero slippage, zero partial fills, and zero settlement disputes; only pmWdw introduces a realistic-execution noise floor. *Second*, every panel uses the per-trade $\sqrt{n}$ Sharpe formula whose monotone-in-n behaviour is documented in the panel tooltip itself. Reviewer trust in the dashboard's headline number ("rising Sharpe") therefore requires reading the formula disclaimer first; the disclaimer is in the tooltip, not in the top-of-page hero card. The asymmetry—striking number above the fold, caveat under hover—is itself an instance of the legibility-vs-deployability gap this paper documents.

AI and Tool Assistance Disclosure

Anthropic Claude, OpenAI Codex, and OpenAI ChatGPT were used as adversarial research and editorial tools for critique, repository inspection, evidence extraction, statistical recomputation support, and draft revision. Abel AI Lab reviewed the outputs and accepts responsibility for the manuscript, evidence, statistics, citations, and claims. These tools are not authors.

Reproducibility Appendix

Primary repository: Abel AI Lab `boundaries-of-retail-prediction-market-trading`. Primary data files include:

- `data/cross-campaign-aggregate.json`
- `data/per-campaign-summary.csv`

- `data/anc/alternative-routes.jsonl`
- `data/anc/attack-registry.jsonl`
- `data/anc/quote-bank.jsonl`
- `docs/references.bib`

The public artifact records 17 raw campaign directories, 13 active campaigns, 156 instrumented rounds, 225 recovered route decisions, 52 strict attack rows, and 40 quote-bank rows.

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